ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

BASIC PRINCIPLES OF COMPUTER PROGRAMMING AND COMPUTER LITERACY

PAPER 2 · PRACTICE D · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 2 Basic Principles of Computer Programming and Computer Literacy paper 2 · practice d.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Programs are not marked strictly on syntax: give marks for correct logic, variables and output. Accept other names, prompts and correct ways of solving the problem.

QUESTION 1

1.1 Project created and named ✓ (1)

1.2 Suitable sprite ✓ (1)

1.3 Suitable backdrop ✓ (1)

1.4 Variables plastic, glass, paper, points (and badge) created ✓✓✓✓ and set to 0 ✓✓✓✓ (8)

1.5 Three ask blocks ✓✓✓, each answer stored in the correct variable ✓✓✓ (6)

1.6 set points to ✓ (plastic × 3) ✓ + (glass × 2) ✓ + paper ✓ (4)

1.7 if points > 49 ✓ then GOLD ✓; else ✓ if points > 24 ✓✓ then SILVER ✓; else ✓ BRONZE ✓ (8)

1.8 say the points ✓ and the badge ✓ joined with text ✓ (3)

1.9 48 points – SILVER ✓; 60 points – GOLD ✓; 16 points – BRONZE ✓ (3)

```
when green flag clicked
set [plastic] to 0
set [glass] to 0
set [paper] to 0
set [points] to 0
ask [How many kg of plastic?] and wait
set [plastic] to (answer)
ask [How many kg of glass?] and wait
set [glass] to (answer)
ask [How many kg of paper?] and wait
set [paper] to (answer)
set [points] to (((plastic) * 3) + (((glass) * 2) + (paper)))
if < (points) > 49 > then
    set [badge] to [GOLD]
else
    if < (points) > 24 > then
        set [badge] to [SILVER]
    else
        set [badge] to [BRONZE]
say (join (join [Points: ] (points)) (join [  Badge: ] (badge))) for 3 seconds
```

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QUESTION 2

2.1 File named correctly with a comment on the first line ✓ (1)

2.2 import turtle ✓; builder = turtle.Turtle() ✓ (2)

2.3 builder.speed(0) ✓✓ (2)

2.4 builder.pensize(2) ✓✓ (2)

2.5 Outer loop with range(8) ✓✓; square drawn inside it ✓; moves 40 px right before the next square (penup / forward / pendown or setx) ✓✓ (5)

2.6 Inner loop range(4) ✓✓; forward(40) ✓; left or right 90 ✓; correct indentation ✓ (5)

2.7 if i % 2 == 0 ✓✓ red, else blue ✓; begin_fill() and end_fill() around each square ✓✓ (5)

```
# YourName - Question 2
import turtle

builder = turtle.Turtle()
builder.speed(0)
builder.pensize(2)

for i in range(8):                  # eight squares
    if i % 2 == 0:
        builder.fillcolor("red")
    else:
        builder.fillcolor("blue")
    builder.begin_fill()
    for side in range(4):           # one square
        builder.forward(40)
        builder.left(90)
    builder.end_fill()
    builder.penup()
    builder.forward(40)             # move to the next square
    builder.pendown()

turtle.done()
```

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QUESTION 3

3.1 File created and named ✓ (1)

3.2 total_fees = 0 ✓✓ outside the function (2)

3.3 def register_learner(): ✓✓; name input ✓; distance as a number ✓; sibling input ✓; global total_fees ✓ (6)

3.4 if distance < 5 ✓ fee 0 ✓; elif distance <= 15 ✓✓ fee 150 ✓; else ✓ fee 250 ✓; correct structure ✓ (8)

3.5 if sibling == "yes" ✓✓ fee × 0.8 ✓✓ (4)

3.6 Fee displayed ✓✓ with 2 decimals ✓ and the name ✓ (4)

3.7 while loop ✓✓; asks "Register another learner?" ✓✓; stops on "no" ✓✓; correct indentation ✓ (7)

3.8 total_fees = total_fees + fee ✓✓ inside the function with global ✓✓ (4)

3.9 register_learner() called in the loop ✓✓; total displayed after the loop ✓✓ (4)

3.10 R150.00, R200.00 and total R350.00 ✓✓ (2)

3.11 Saved and printed / presented ✓ (1)

```
# YourName - Question 3
total_fees = 0
```

```
def register_learner():
    global total_fees
    name = input("Learner name: ")
    distance = float(input("Distance from school (km): "))
    sibling = input("Sibling on the bus? (yes/no): ").lower()

    if distance < 5:
        fee = 0
    elif distance <= 15:
        fee = 150
    else:
        fee = 250

    note = ""
    if sibling == "yes":
        fee = fee * 0.8
        note = " (20% sibling discount)"

    print(f"{name} pays R{fee:.2f} per month{note}")
    total_fees = total_fees + fee

again = "yes"
while again == "yes":
    register_learner()
    again = input("Register another learner? (yes/no): ").lower()

print(f"Total monthly fees: R{total_fees:.2f}")
```

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TOTAL: 100

END OF MARKING GUIDELINE